

Credit Name:CSE 2140 2nd Language Programming
Assignment Name: Chapter 4 Mastery

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I approached this project by forming an IPO chart , for the Math Tutor application it meant figuring out that I need to implement random numbers and the user's response. While, for the Delivery Service application it meant forming inequality equations in order to establish value and weight thresholds.

A part which I found confusing was how to randomly generate an operator, however after reviewing the textbook and the previous hurricane skillbuilder, I got the idea to have the program generate a random number which corresponded to the case which held a operator.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

How did you decide on the tools, data structures, or algorithms to use?

For the delivery service assignment, I had written all the outcomes the program could output in order to produce an accurate response to the user's inputs. For the math tutor by figuring out the outcomes I had realized with the division operator, the answer could be a decimal, so I casted (double) on the equations.

I utilised the switch for the Math tutor as it could create cases based on a randomly generated integer value. I used if and else statements for the delivery service, because the output message was dependant on overlapping ranges. E.g "Too heavy and Too big" and "Too heavy" overlap as both have the weight value as greater than 27.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

How did you test your solution along the way to make sure it was working?

For the Math Tutor, I had tested the application only using case 0 (addition) at first to resolve any errors. I had also only tested the parameters of weight at first before developing the dimensions of package.

I tested the code for Math Tutor, by inputting completely wrong values, values that were a decimal off, and correct values to check the code. I had also done this for the Delivery Service to test the output messages.

Overcoming Challenges

What part of the problem was the most difficult for you?

How did you handle moments when you felt stuck or unsure of what to do next?

Developing the if and else statements for the Math Tutor application was especially hard as I kept running into problems of incorrect format and assignment of variables.

I instead focused on developing the statement for case 0, which allowed me to deduce and resolve errors.

Learning

Was there anything you learned that you think will help you with future challenges?

I learned how to work around overlapping conditions by forming separate if and else statements. I had also learned how to utilise random number generation has many more uses than simply a number, rather operators, outcomes etc.